

Read, Replace, Never Rewrite: Continuous Thoughts Carry Search Structure, Not Answer Identity

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Abstract

Models can sidestep natural language and reason through latent chain-of-thought by bypassing the token-generation phase and feeding the last-layer hidden states back into the input. Because the thought does not commit to a single token per pass, it can represent multiple candidate tokens at once. The superposition hypothesis sees latent thoughts as a weighted sum of search branches, an interpretation supported so far only by theoretical construction and evidence that the branches are readable from the vector. Since readable components are not necessarily used in computation, we run the first causal, pre-registered test using the theory’s own setting, a two-layer model trained from scratch for graph-reachability tasks. If the thought is such a sum, branch arithmetic should be possible, so we test it with three causal interventions: reading, rewriting, and replacing the thought. It can be read: the search frontier is decodable (AUC 0.9979) and the thought is necessary (−42.9 points). It cannot be rewritten: editing a branch in every pre-specified basis, or even under maximal linear-subspace removal, has no effect at any intermediate step. It can be replaced, but not always: transplanting entire thoughts redirects the answer to the donor’s when the problem differs in structure (−91.7 points), and does nothing when it differs only in which candidate is correct, a pre-registered rejection. The single edit that works is an identity swap at the final step, which completely flips the answer. Structure is carried, while identity is bound late.

1 Introduction

Chain-of-thought (CoT) improves accuracy on complex tasks, and because the intermediate steps are written in natural language, humans can follow the process towards the answer (Wei et al., 2022). COCONUT (Hao et al., 2024) introduces latent reasoning through a continuous chain-of-thought, entirely bypassing token generation, and gives that

readability up. It works by feeding back the last hidden state as the next input embedding, so at a latent step the model never commits to a single token, and the latent CoT can reach conclusions that no single forward pass could. During latent steps, the reasoning lives only in vector space, and the only way to read it is with interpretability tools.

On graph reasoning tasks, an internal map resembling BFS emerges during the latent phases (Zhu et al., 2025b). The leading mechanistic interpretation of it is superposition: multiple tokens (nodes) get carried forward through a search frontier, and the thought becomes a weighted sum of the search branches it holds at once. This superposition phenomenon is supported by a theoretical construction showing that a small transformer can solve the task in this way (Zhu et al., 2025b). It is also backed by readouts: inner products between the thought and the node embeddings separate the nodes by their search status in trained models (Zhu et al., 2025b,a). An empirical survey found this observable pattern only in small models trained from scratch, as fine-tuned larger models find shortcuts and bypass their latent tokens (Rizvi-Martel et al., 2026), so the phenomenon is real but lives in a narrow regime. Every kind of evidence observes the thought, but none intervenes.

A representation can be readable without the model actually using it in its computation (Belingov, 2022; Hewitt and Liang, 2019). We ask directly: what is actually inside a continuous thought?

If the thought is a sum of branches, arithmetic on those branches should be possible and have predictable consequences: if we remove the branch carrying the answer, the model should lose its answer. We test it on the theory’s own base model, a two-layer transformer trained from scratch on graph reachability, in three ways, each addressing the gap the previous one leaves open.

Can the thought be *read*? Yes. The search fron-

tier is decoded almost perfectly (AUC 0.9979), the BFS wave appears exactly as predicted at every latent step, and ablating the thought causes a drop of -42.9 points. The thought is readable and necessary.

Can it be *rewritten*? No. Per-branch edits have no causal effect at any intermediate step, in any of the bases we pre-specified (a pre-registered rejection), and neither does removing every direction from which the frontier can still be linearly read.

If not, can it be *replaced*? Only partly. Transplanted thoughts switch the answer to the donor’s correct one when the donor’s problem differs in structure (-91.7 points), and do nothing when the donor’s problem differs only in which candidate is the target output.

The single edit that works is an identity swap at the final step, which completely flips the answer. Structure is carried, while identity is bound late.

Studies on causal intervention leave open the choice of basis, thresholds, and when to stop. This freedom could threaten validity, and because two of our results are negative, those areas are exactly where we are the most rigorous in our protocol and criteria. We pre-registered the confirmatory analysis, decision criteria, and encoding before data acquisition began, and we documented every amendment. Both hypotheses that failed to qualify are reported as findings.

Our contributions are as follows:

- **The first causal test of the superposition theory.** We pre-register it and run it on the theory’s own base model. The observational readouts fully replicate, but the causal-linear reading is rejected: readable does not mean used in computation (Secs. 4, 5).
- **What the thought is.** The structure is causally carried through a transplantable search state, but the identity remains readable-only mid-reasoning, and it is bound to a token at the readout (Sec. 6).
- **Where interventions’ effects land.** The one linear edit with any effect is a final-step identity swap, which flips the answer completely. Interventions that aim to switch the winning candidate belong at the final, binding step (Sec. 5.4).
- **Methods.** A causal-Jacobian basis adapted for latent CoT, and amnesic probing used to

validate a negative result instead of a positive one (Sec. 5, App. E).

- **Reproducibility.** We correct the reference implementation’s unseeded serialization, and we fully release the pipeline: source files, thresholds, amendments. Every reported number is traceable to a committed file (App. C, F).

2 Background and Framework

2.1 Latent reasoning

In discrete Chain of Thought, each reasoning step produces a single token (Wei et al., 2022). By committing to the respective token, other viable options remain unexplored, so context and possibly relevant information are lost. Latent CoT (Hao et al., 2024) attempts to mitigate this by bypassing the unembedding step and introducing the hidden state vectors produced by the last layer directly into the input embeddings for the corresponding next step, so no intermediary tokens are produced. Therefore, at a given latent step, the final-layer hidden state defines the thought $\mathbf{z}_k \in \mathbb{R}^d$, and constitutes the input embedding of the next latent step. Related latent-reasoning designs include distillation-based continuous thoughts (Shen et al., 2025) and recurrent-depth architectures (Geiping et al., 2025). We study the COCONUT form throughout.

$$\mathbf{e}_{(k+1)}^{\text{in}} := \mathbf{z}_k, \quad k = 1, \dots, K \quad (1)$$

After K continuous latent steps, the answer is produced at the readout position. The thought at step k is the latent state every experiment in this paper reads, rewrites, or replaces.

2.2 Substrate: task, model, training

This paper’s evaluation set is comprised of graph-reachability problems, following the theory’s settings. Consider a directed acyclic graph $G = (V, E)$ whose nodes are assigned dedicated tokens in the model’s vocabulary, serialized as a list of edges. Given two candidate nodes, the query asks which one is reachable from the root. The answer is a single token, the one assigned to the correct candidate. While traditional chain-of-thought explicitly lists the path between nodes, the latent version trains the model to progressively replace these discrete steps with hidden representations.

Because the nodes’ tokens are not natural-language tokens, their embeddings are learned only

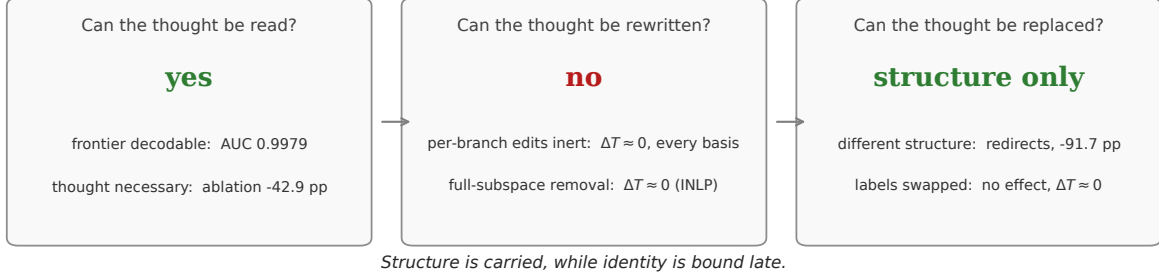


Figure 1: The three causal tests and their outcomes: READ: the search frontier is decodable (AUC 0.9979) and the thought is necessary (−42.9). REWRITE: interventions applied to branches have no causal effect in any basis we pre-specified, a pre-registered rejection. REPLACE: transplant replacements change the output when the donor differs in structure (−91.7) and do nothing when only the candidates are swapped. Structure is carried, while identity is bound late.

from the task. The input-embedding basis of §5 therefore carries task-related search roles only, with no external meaning that could distort the interventions.

The model is a two-layer GPT-2 decoder with eight attention heads and 768 dimensions. It is trained from scratch, per [Zhu et al. \(2025b\)](#)’s specifications. Training occurs over 300 epochs across different stages: at stage m (epochs $25m$ to $25m+24$, with a 0.1 stage mixing factor), the first m discrete reasoning steps are replaced by latent continuous ones. Unlike the original fixed-length training, we run the final full-task stage under early stopping on validation accuracy (a documented protocol amendment). All experiments are run on a bit-exact, consumer-hardware reimplementation of the original training and inference code, with verifications on logits, loss, and gradients (Appendix C).

We adopt the exact architecture, task, and scale of [Zhu et al. \(2025b\)](#), the setting where both the superposition construction and its inner-product readout were established. Independent external analysis ([Rizvi-Martel et al., 2026](#)) shows that this superposition phenomenon occurs only in small models trained from scratch, among the regimes tested. Fine-tuned larger models do not materialize the search map the same and find shortcuts instead, achieving accuracies up to 96.6% with the latent tokens removed entirely. The regime studied here is one where the hypothesis is live. This is why a causal test in these settings is viable: if the thought arithmetic theory fails even on its home ground, the experimental setup cannot be the one to blame.

2.3 The superposition hypothesis

If the latent continuous steps build a BFS-like structure, where the thought is a weighted sum of the node directions at a given latent step, then branch arithmetic should be possible and causally relevant.

$$\mathbf{z}_k \approx \sum_{v \in \mathcal{F}_k} \alpha_{k,v} \mathbf{u}_v, \quad \alpha_{k,v} > 0, \quad (2)$$

First, we explore whether the components are readable: their inner products $\langle \mathbf{z}_k, \mathbf{u}_v \rangle$ should separate nodes by their search status, forming the observable frontier. By search status, at a chosen latent step, a node is either not yet reachable, already reached, on the current frontier, or on the answer path. Second, we explore whether the components are writable, meaning that editing the sum component-by-component should be causally relevant. For example, if we remove $\alpha_{k,v} \mathbf{u}_v$, branch v should no longer be part of the search.

[Zhu et al. \(2025b\)](#)’s construction is an existence proof, but trained models may implement the frontier differently, and whether the readable pattern is the causal encoding is this paper’s main question.

2.4 Terminology

We refer to superposition as the ability of the hidden state to simultaneously encode multiple reachable nodes, not committing to a single one per step. This set of encoded nodes is the search frontier, and the encoding is exactly the weighted sum of Eq. 2. It is distinct from the superposition of more features than dimensions ([Elhage et al., 2022](#)), where features are represented non-orthogonally.

3 Experimental Design

3.1 Measurements

We present the following definitions:

- T — the target readout: a score between 0–100, calculated from the answer distribution over the two candidates, for a single graph. $T = 100 \cdot p(\text{target}) / (p(\text{target}) + p(\text{decoy}))$
- ΔT — an intervention’s effect on a graph: $\Delta T = T_{\text{intervened}} - T_{\text{baseline}}$
- e — the non-candidate answer probability: separating *redirection* (large $|\Delta T|$ with $e \approx 0$: the model flips the answer) from *destruction* ($e \rightarrow 1$: the model leaves the task’s answer space). We place strict limits on this variable so that broken reasoning cannot be mistaken as a successful redirection.

An intervention’s effect on a graph is strongly concentrated at the extremes: near 0 (the answer remains the target) or near -100 (it switches to the decoy), with few cases in between (Appendix D). A median can therefore fall in a gap that no single graph covers, so we look at both the median and the percentage of graphs with $\Delta T \leq -50$. Each condition runs on $n = 100$ pre-set problems with a single correct answer, in which both candidates are present (Appendix A). If the outcomes are too close to a threshold, we further analyze a pre-established number of new graphs.

3.2 Protocol

Causal intervention papers inherently give researchers much freedom to tweak variables such as decomposition basis, effect threshold, and renormalization policy (Gelman and Loken, 2013; Simmons et al., 2011). This heightens the risk of tuning these choices after seeing results to reach a desired conclusion. Because this study was designed to rigorously test an existing theory, negative outcomes were always on the table. To ensure that a null result would count as true evidence, the exact thresholds were locked in before data collection began. Additionally, the core hypothesis was easy to test before committing to any intensive follow-ups, so we pre-registered the initial tests and shaped the experimental program around their outcomes.

To ensure our results reflect the phenomenon instead of our experimental design choices, we followed a strict protocol presented in Table 1.

Amendments correct operations, never criteria, and are documented in Appendix B.

While pinning the serialization, we found that the base pipeline shuffled each problem’s components using an unseeded global generator, so no two runs were seeing the same serialization. We corrected this by pinning an order per problem and report the process in Appendix C.

4 Reading the Thought

Can the model’s internal search map be read directly from the latent thought? We answer in three steps: we replicate the theory’s setting and the inner-product pattern between thoughts and node embeddings (the readout), measure how well the search frontier can be decoded from the thought with linear probes, and test whether the thought carries any causally necessary information at all by ablating it.

4.1 Replicating the structural baseline

We recreate Zhu et al. (2025b)’s exact setting as described in §2.2: a two-layer transformer trained from scratch on graph reachability, reasoning through K continuous thought steps. We define the readout score as the inner product between the thought at latent step k and node v ’s token embedding.

$$s_k(v) = \langle \mathbf{z}_k, \mathbf{u}_v \rangle, \quad (3)$$

A BFS from the root classifies every node at step k as not reachable ($\mathcal{N}$), reached at an earlier step ($\mathcal{R}_k$), on the current frontier ($\mathcal{F}_k$), or on the optimal path ($\mathcal{O}_k$).

The theory predicts the ordering:

$$\begin{aligned} \text{med}_{v \in \mathcal{N}} s_k(v) &< \text{med}_{v \in \mathcal{R}_k} s_k(v) \\ &< \text{med}_{v \in \mathcal{F}_k} s_k(v) \\ &< \text{med}_{v \in \mathcal{O}_k} s_k(v) \quad (\forall k) \end{aligned} \quad (4)$$

A model qualifies for our analysis only if the ordering of Eq. 4 holds at every step, and its accuracy on held-out problems is $\geq 93\%$. The primary model qualifies, solving 94.5% of held-out problems with the ordering correct at every latent step.

We observe that reachable nodes are activated in the thought vector in natural search order, and the frontier branches out at each step. This holds in all four independently trained models, including the two that failed to reach the pre-specified acceptance criteria, so the ordering is not the property of one lucky model.

Safeguard	Prevents	Incorporation
Pre-specified, locked decision criteria	<i>post-hoc threshold tuning; uninterpretable nulls</i>	All thresholds were committed before any corresponding data existed (commit hashes: App. A), with possible failures specified in advance (e.g., the transplant test of Sec. 6.2 required median $\Delta T \leq -30$, half of the graphs at $\Delta T \leq -50$, and median non-candidate answer probability ≤ 0.05).
Step-by-step hypothesis testing	<i>flexibility interpreted as confirmation</i>	Each confirmatory test was designed only after the previous verdict, with the criteria fixed before we had access to data, and exploratory analyses are labeled as such and cannot revise verdicts.
Positive and negative controls	<i>measurement errors, model failures disguised as null results</i>	Every run carried a null intervention (self-transplant, must be 0.0), matched random directions that must result in nulls, and a positive sensitivity control that must show a large effect (per-step zeroing, -42.9 on the same graphs).
Experimental setup validation, fixed seed budget	<i>cherry-picking training runs</i>	Confirmatory analyses only on models with $\geq 93\%$ held-out accuracy and readout ordering at all latent steps. The misses were reported, but never analyzed, and no seeds added after outcomes were observed (2/4 passed).
Pinned graph to text serialization	<i>counterfactual pairs differing by accident</i>	Fixed per-problem serialization order, so matched donors differ only where intended.

Table 1: Protocol safeguards, the failure mode each addresses, and how they were used in this study. Protocol amendments are documented in App. B.

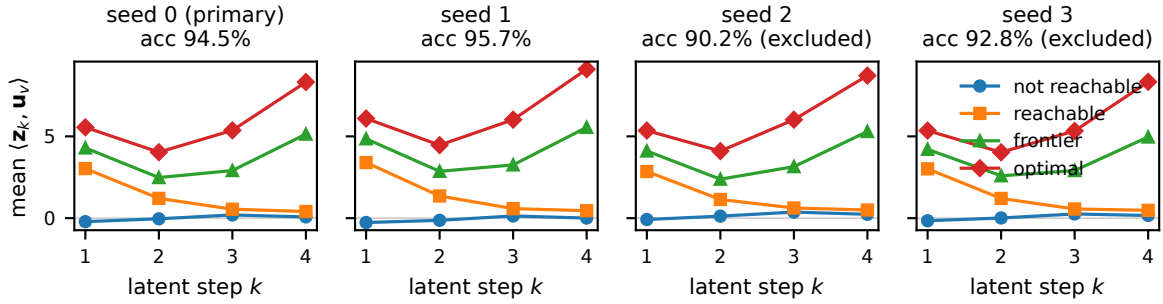


Figure 2: The structural readout replicates in all four independently trained models. The median readout score (Eq. 3) by search status at each latent step, and the ordering of Eq. 4 is correct at each step, including in the two models that miss the accuracy threshold.

The search structure is what stays consistent, while the conversion of the target’s reachability status into an answer varies.

4.2 Linear decodability of the frontier

We find that the nodes in the current search frontier can be read out using linear probes fitted on training data, reaching AUC 0.9979. Every result is measured on held-out problems that were never used for fitting.

We employ the following linear probe:

$$\hat{p}_k(v) = \sigma(\mathbf{w}_v^\top \mathbf{z}_k + b_v), \quad \text{AUC over } \{(G, k, v)\} \quad (5)$$

The directions $\{\mathbf{w}_v\}$ reappear in §5 as the probe basis.

4.3 The thought is causally necessary

We recall the sensitivity control from §3.2, here as an experiment. We zero out the entire thought to test whether it contributes relevant information.

This is the general intervention format:

$$\mathbf{z}_k \leftarrow \tilde{\mathbf{z}}_k \quad \text{for } k \in \mathcal{K} \subseteq \{1, \dots, K\}, \quad (6)$$

For total ablation, $\tilde{\mathbf{z}}_k = \mathbf{0}$, $\mathcal{K} = \{1, \dots, K\}$. The prompt, weights, and readout remain the same.

Effects are measured per graph as ΔT (§3.1), reported as the median over the pre-specified $n = 100$ evaluation problems. Zeroing the thought at every step drops performance by -42.9 points, so the thought is causally necessary for task performance. The effect replicates on held-out problems (-25.4 , not as strong, but still detectable) and on seed-1 (-36.8). Because this ablation produces a

Model	Held-out acc.	Accepted	Wave
seed 0 (primary)	94.5	pass	all K steps
seed 1	95.7	pass	all K steps
seed 2	90.2	<i>fail</i>	all K steps
seed 3	92.8	<i>fail</i>	all K steps

Table 2: Answer accuracy is dependent on the training seed (within 90.2–95.7, with the pre-specified threshold inside it). The structural readout replicates in all four models, including both that miss the accuracy threshold. Per protocol (Sec. 3), models that did not pass are reported but excluded from confirmatory analyses, and no more seeds were drawn after the outcomes were observed.

large, reliable effect on these graphs, it will serve as the positive sensitivity control for every analysis onward, as it shows the instrument can detect a real effect when one exists.

By recreating the setting of the theory paper, decoding the latent thought’s components with linear probes, and ablating the thought channel, we have established that the thought is readable (AUC 0.9979, Eq. 5) and necessary (-42.9 , Eq. 6). However, the necessity of the whole thought is not evidence that the readable content is what is used in computation. Nothing so far identifies which nodes, if any, carry the answer. If the readable superposition is the internal working memory, its components should be rewritable.

5 Rewriting the Thought

Can components of the thought be edited? If the thought is a sum of branches, more specifically a weighted sum of frontier nodes, arithmetic on those components should have predictable consequences. Removing the branch carrying the answer should make the model lose the answer, collapsing on a single branch should commit the search to it, and adding a frontier node should insert its corresponding branch into the search.

SUBTRACT(v) :

$$\tilde{\mathbf{z}} = \|\mathbf{z}\| \cdot \text{normalize}(\mathbf{z} - \langle \mathbf{z}, \hat{\mathbf{u}}_v \rangle \hat{\mathbf{u}}_v),$$

COLLAPSE(v) :

$$\tilde{\mathbf{z}} = \text{sign}(\langle \mathbf{z}, \hat{\mathbf{u}}_v \rangle) \|\mathbf{z}\| \hat{\mathbf{u}}_v, \quad (7)$$

INJECT(x) :

$$\tilde{\mathbf{z}} = \|\mathbf{z}\| \cdot \text{normalize}(\mathbf{z} + \gamma \hat{\mathbf{u}}_x),$$

$$\gamma = \text{med}_{f \in \mathcal{F}_k} \langle \mathbf{z}, \hat{\mathbf{u}}_f \rangle,$$

Each operation acts along a node direction drawn from a basis, and both the basis and the thresholds it

must pass are pre-established (§5.1). The outcome is uniform: no intervention applied in any basis, at any step, has a detectable behavioral effect. This cannot be an instrument malfunction, as the same instrument registers -42.9 from total ablation.

5.1 Operations, bases, and qualification thresholds

All three aforementioned operations act through the intervention operator of Eq. 6, per Eq. 7, and differ only in how they modify the thought. Each edited vector is rescaled to the pre-intervention norm of that thought, so the operation changes only the direction of that component, not its magnitude. Each intervention is paired with two control tests: the same operation along a random unit direction of matched magnitude, or along a non-frontier node’s direction. The operations are applied at the step where the target branch’s root enters the frontier and, in follow-up experiments, are also reapplied at every subsequent step.

Because directions are not predefined in the thought, we assign each node one unit direction $\hat{\mathbf{u}}_v$, and the full assignment forms a basis. We pre-specified three assignments: (1) the input embedding basis $\hat{\mathbf{u}}_v = \mathbf{u}_v / \|\mathbf{u}_v\|$ (the theory’s own alphabet), since the readout (Eq. 3) decodes the frontier from these exact directions, (2) the unembedding basis, which coincides with (1) on this model, and (3) the probe basis, the linear probe directions from which the frontier is best read. After exhausting this sequence, we added a new exploratory basis: a causal-Jacobian basis where the directions were defined by causal influence on the readout, instead of embedding structure or decodability (construction in Appendix E). Being marked as exploratory, it cannot affect the registered outcome.

An operation is evaluated only on a basis where editing the thought demonstrably changes the answer, which serves as a qualification threshold: SUBTRACT on the answer branch has to move the median ΔT by at least -25 , while the effects caused by the matched random intervention stay within a ± 10 range. If SUBTRACT fails, COLLAPSE or INJECT are not evaluated, as they operate in the same setting, so a null on them would be uninterpretable. The threshold is what makes an eventual negative result evidence instead of absence.

Basis	SUBTRACT (answer branch)	random control	qualification threshold
input-embedding	−0.0	0.0	fail
unembedding	degenerate (tied embeddings; footnote)		
probe basis	−0.0	0.0	fail

Table 3: No per-node basis demonstrates causal effect: subtracting the answer branch’s component does not have any effect on the readout (pre-specified criteria: median $\Delta T \leq -25$ with control within ± 10 ; $n = 100$). Replacing the unit-norm version of the first row with the norm-preserving update is documented in Appendix B.

Condition (per-step unless noted)	input-emb.	probes	causal-Jacobian [†]
one-shot SUBTRACT ($t=1$)	−0.0	−0.0	0.0
SUBTRACT, same node, every step	−0.0	−0.0	−0.0
eliminate entire answer-carrying path, excl. final step	−0.01	−0.0	−0.0
random-direction control	−0.0	−0.0	−0.0
eliminate entire answer-carrying path, incl. final step	−65.1	—	−99.5
total ablation (sensitivity control, §4)		−42.9	

Table 4: Escalated deletion has no effect mid-reasoning. Median ΔT ($n = 100$), even removing every node of the answer-carrying path at every step leaves behavior mostly unchanged, unless the final step is included (the readout), foreshadowing §5.4. [†]post-verdict exploratory basis.

5.2 No basis passes; per-branch subtraction is inert

We recall that total ablation on the same graphs, measured with the same instrument, reported -42.9 . Against this baseline, no basis passes the qualification threshold (Table 3).

Per the pre-specified thresholds, the thought is not a sum of branches that lead to a causal effect when they are linearly modified. Because no basis qualified, COLLAPSE and INJECT were not evaluated. To investigate the lack of causality more than what the pre-registered tests allowed us, we escalated the deletion in two documented subsequent ways (Appendix B): by reapplying the subtraction at every latent step or by eliminating all of the nodes of the answer-carrying path at every step.

If we apply maximal linear deletion by removing every component of the answer-carrying path at every step excluding the final one, the median ΔT drops by -0.0 .

5.3 The null is not under-deletion

Single-direction deletion can miss directions that encode the same information, so the readable content may survive the projection. To mitigate this, we employ amnesic probing:

$$\tilde{\mathbf{z}} = \mathbf{P}^{(m)} \mathbf{z}, \quad \mathbf{P}^{(m)} = \prod_{i=1}^m \left(\mathbf{I} - \hat{\mathbf{w}}_i \hat{\mathbf{w}}_i^\top \right), \quad (8)$$

This is an iterated null-space projection per node. We refit the frontier probe after each projection,

remove the targeted direction, and repeat until the newly fitted probe can no longer decode the frontier from the projected thought ($\text{AUC} \leq 0.55$) or an iteration limit of 40 is reached.

If we remove every direction from which the frontier can still be linearly read, the answer does not change: median $\Delta T = -0.0$, $T_{\text{post}} = 99.99$. Removing the same number of random directions of the same norm does exactly as much (0.0), so by the amnesic-probing test, the linearly readable frontier content is not causally used by the computation. After removal, what a probe can read is nearly a coin flip: $\text{AUC } 0.5\text{--}0.67$, versus 0.50 at chance. The null is not caused by our measurement, as the same instrument recorded -42.9 in Section 4, and larger effects are reported ahead (§6).

5.4 Identity is editable only at the readout

At the final step, switching the target candidate with its matched decoy direction in the exploratory causal-Jacobian basis changes the answer almost completely (-99.97 and 0.0001). The identical swap applied at every intermediate step does nothing, and the effect only happens at the readout. We return to this phenomenon in Section 6.3.

We have established that the thought is readable, necessary, and not rewritable per component or branch, even if we apply full subspace deletion in any of our selected bases, including the exploratory one. This reflects a separation between the computation and the readable aspect of the thought, as the readable superposition is not the causal encoding.

If its content cannot be rewritten, are the thought’s components even causal at all? We stop editing and replace the entire state.

6 Replacing the Thought

We stop rewriting and start replacing the thought altogether. If the thought’s components cannot be rewritten linearly, two questions remain. Does the thought carry causal content that our dictionaries simply cannot express? Or is it a passive copy: readable, necessary as a whole, but causally inert component-by-component? One way to separate them is to substitute the entire search frontier: we replace the original thought mid-reasoning with one from a donor and see whose search the model completes. There are two cases: (1) donors with entirely different problems, (2) donors with an identical problem in which only the two candidates’ labels are swapped. The main finding of this experiment is how different the donor must be for the transplant to take effect.

6.1 Transplants of entire thoughts

After the rejection of §5 (Appendix B), we substitute entire thoughts instead of editing their components as documented subsequent exploratory analyses. A transplant is an intervention operation with the donor’s thoughts as the source:

$$\tilde{\mathbf{z}}_k = \mathbf{z}_k^{\text{donor}}, \quad k \in \mathcal{K}, \quad (9)$$

We obtain these thoughts by extracting them from the cache of an independent run on the donor’s own problem, and then simply substituting them into the original reasoning steps. We do not interfere with the base prompt, so the baseline graph remains the same. We define a matched donor as a different graph built on the original’s serialization, where only the graph content differs, while a random donor is represented by thoughts from a randomly chosen other problem. We also introduce some control tests in §3.2 for the transplants: self-transplant, baseline reserialization to confirm its neutrality, and self-transplant from self-reserialization to verify pipeline integrity.

While the matched donor steers the answer cleanly, the random donor does not redirect, so the effect is driven by the transfer of reasoning, not by the model being fragile to external interference. Transplants applied at a single step almost fully recover through recalculation from the original prompt (−2.9 versus sustained −91.7). The in-

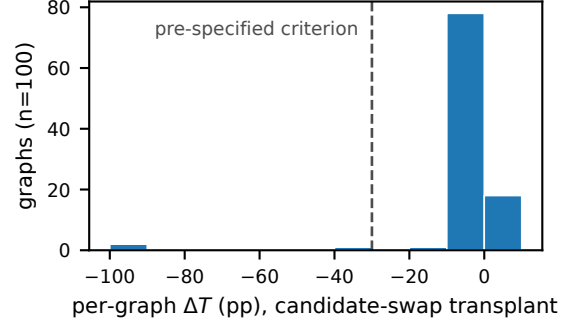


Figure 3: Per-graph effect of the label-swap transplant ($n = 100$). The zero result is far from the pre-specified criteria (replications in Appendix D).

ternal search map is continuously updated and easily reconstructed, so latent thoughts act as the primary carrier under sustained substitution, not as the only instance of the search path. But transplanting the thought at the final step alone flips the answer almost completely (−99.95), a phenomenon we return to in §6.3. If the thought carries the winning candidate, we make the following pre-registered prediction: a donor identical except for the two candidates’ labels should reverse the answer.

6.2 The label-swap transplant (pre-registered)

We build the donor problem from the original graph with the two candidate labels swapped:

$$G^{\text{swap}} = \pi_{T \leftrightarrow D}(G), \quad (10)$$

where $\pi_{T \leftrightarrow D}$ interchanges the candidates’ labels everywhere in the serialization. Topology, the depths, and reachability of nodes of the two graphs remain the same, but the correct answer becomes D . We lock the problem formatting to the original one (§3.2) so that candidate identity remains the only difference. We verify the baseline requirements for every pair, and any pairs that fail to meet the criteria get rejected by default (Appendix A). The swap donors obtain the opposite answer on their own problems (median $T(\text{answer}=D) = 99.8$, and 79% correct), so a null result would be a finding, not a bug: a genuine thought is what gets transplanted.

The hypothesis that the carried state encodes the candidate’s identity is rejected per pre-specified criteria. The median effect is −0.0 against a −30 threshold, and 0.02 of transplants flipping against 0.5 cannot be categorized as a marginal miss.

Blending in thoughts from the donor at $\alpha \in \{0, .25, .5, .75, 1\}$ in a norm-preserving manner re-

Condition	n	median ΔT	median non-candidate probability
baseline reserialization	100	−0.0	0.0001
self-transplant	100	0.0	0.0001
donor, intermediate steps	96	−91.7	0.0019
donor, first step only	96	−2.9	0.0018
donor, final step only	96	−99.95	0.0001
donor, all steps	96	−99.97	0.0001
random donor, intermediate	100	−30.4	0.98
total ablation (baseline, Sec. 4)	100	−42.9	0.76

Table 5: Entire thought transplants (subsequent exploration, Appendix B). A matched donor’s mid-reasoning thoughts redirect the output, with the answer mostly being the candidate (−91.7 or the non-candidate 0.0019). A random donor’s thoughts do not redirect but derail it (0.98). The non-candidate column is what separates genuine carriage from fragility. $n=96$ rows: four problems lack a matched donor (Appendix D).

	criterion	primary	ind. seed	held-out [‡]
median ΔT (candidate swap)	≤ -30	−0.0	−0.0	−0.01
fraction $\Delta T \leq -50$	≥ 0.5	0.02	0.0	0.08
median T_{post}	≤ 60	99.98	99.98	99.91
median non-candidate probability	≤ 0.05	0.0001	0.0001	0.0002
placebo swap (control)	$ \cdot \leq 10$	0.0	0.0	0.0
self-transplant (control)	≈ 0	0.0	−0.0	−0.0
ablation baseline (sensitivity)	detects	−42.9	−36.8	−25.4

Table 6: The pre-registered label-swap transplant fails its criteria on the primary model and both replications, with every control passing. A donor that verifiably computes the opposite answer (99.8 on its own problems), transplanted at every intermediate step, leaves the original answer unchanged. [‡]Replication on the held-out split is not a mandatory condition for our main claims, as its baseline effect is attenuated (−25.4), and borderline results trigger a pre-specified data requirement (Appendix D).

sults in a flat curve. This is expected, since the endpoints behave identically, and the $\alpha=1$ endpoint has the same results as the ones from the transplant (difference 0.0), validating our implementation. Our pre-specified trajectory analysis confirms that 99/100 graphs remain completely unresponsive to the blend.

6.3 Resolution: structure is carried, while identity is bound late

The donor’s thoughts either differ from the original ones and this difference is causally inert, or the donor’s thoughts barely differ from the original, so the swap does not occur in the thought. We separate these with the following measurement:

$$c_k = \cos(\mathbf{z}_k^{\text{own}}, \mathbf{z}_k^{\text{donor}}), \quad k \text{ intermediate}, \quad (11)$$

computed with all donors aligned with the original serialization, so only the problem’s content differs.

The candidate swap barely changes the thought (0.9975, rel- L_2 0.07). Candidate identity is hardly written into the mid-reasoning state to begin with, so transplanting it is nearly a self-transplant. A swap that preserves the structure, even if it moves

Donor condition	median cos	median ΔT
candidate swap ($T \leftrightarrow D$)	0.9975	−0.0
placebo swap (irrelevant labels)	0.991	0.0
path-interior swap	0.77	−0.08
different graph (baseline, §6.1)	0.34	−91.7

Table 7: Geometry against causality. Swaps that influence identity leave the mid-search thought a near-copy of the original’s own (candidate swap: 0.9975) or disturb it significantly (interior swap: 0.77), and have no causal effect either way. Only structural change (different graph: 0.34) is causal. Causal effect depends on structural difference, not geometric distance.

the thought as far as 0.77, still does nothing (−0.08), while the structural change at 0.34 carries −91.7. Causal effect is therefore a function of what changed, not of how much.

We define the structure as the unlabeled search tree, covering the topology, reachability, and depth of its nodes, and which structural position holds each candidate. On the other hand, the identity is the token assigned to a specific node. The can-

	readable	causally used	evidence
structure, mid-search	yes	yes	wave (Sec. 4); transplant -91.7 (Sec. 6.1)
candidate identity, mid-search	yes	no	AUC 0.9979 (Sec. 4); edits ~ 0 , INLP (Sec. 5); swap-donor null (Sec. 6.2)
candidate identity, at readout	—	yes	final-step swap -99.97 (Sec. 5.4)

Table 8: Mid-reasoning the structure of the thought is both readable and has causal effect, while candidate identity is only readable. Identity becomes causally relevant only at the readout step.

didate swap results in a tiny (0.07) change, concentrated 94 times more than chance (0.5) in the span of the two candidates’ embedding directions (fraction 0.151 versus 0.0016), and we refer to it as the residual. The final-step swap of §5.4 and the final-step-only transplant (Table 5) are the same fact measured by two instruments, showcasing how binding happens at the readout. The identity leaks into the thought for a very small number of graphs, as 0.02 of transplants change the outcome, but this happens at a negligible level.

These experiments show that the thought carries the search’s structure causally, but the identity only readably, as it becomes causal only at readout, where identities are bound back to their tokens.

7 What a Continuous Thought Is

The internal search map built over K continuous latent steps carries the structure of the search: which nodes are reachable, what the frontier contains, which position holds the winning candidate. The assignment of nodes to tokens is resolved only once at readout, through attention to the original prompt. This structure-identity split aligns with the classic tradition of role-filler separation (Smolensky, 1990; Fodor and Pylyshyn, 1988) in the variable-binding problem: roles are carried through the recurrence steps, but binding happens only at the very end. This mechanism also explains the causal-Jacobian asymmetry described in §5.4 and it makes it an expected consequence of this framework.

We have not tested whether this structure/identity split is specific to latent reasoning or if it is tied to this experimental environment, as our empirical findings were established on the theory’s home ground.

8 Related Work

The setting and its theory. Reasoning in the continuous latent space was introduced by COCONUT (Hao et al., 2024), the first paper to report models that simultaneously explore multiple candidate paths through the graph. Zhu et al.

(2025b) provides the existence proof and the observational readout we replicate in full (§4), but the editable sum-of-branches reading of it is rejected after causal testing. Our findings delimit what the construction does and does not establish: a model of this size can solve the graph reachability task using a branch-sum mechanism, but the solution the trained model learns carries the frontier using a different encoding. A follow-up (Zhu et al., 2025a) by the same authors analyses how the superposition readout emerges over the course of training, and while both works analyze the trained model observationally, neither attempts to edit its states. In the case of Gözeten et al. (2025), the superposition is constructed at the input by designing the continuous tokens to be the probability-weighted averages of token embeddings. There, the branch weights are explicit inputs, so removing a branch is just setting a known coefficient to zero, with no decomposition required. We ask whether an editable branch-sum exists at all, and in which directions, only for training-emergent superposition, not constructed superposition.

Observational accounts of latent reasoning.

Rizvi-Martel et al. (2026) investigate and map where superposition patterns appear across training regimes and scales, showing that they appear only in small models trained from scratch, as fine-tuned larger models find a shortcut to bypass their latent tokens, achieving high accuracies nonetheless. The tools they applied are Logit Lens and probing, and their findings support our choice of experimental design and rule out a fine-tuned extension. The question they left open is whether the patterns in the smaller models are causal or not: this is exactly what our paper answers. Probing for safety purposes of continuous-thought models (Ramjee, 2026) detects misaligned reasoning from the latent thoughts, but such monitoring is based on readability alone, and this is exactly the read/use gap our findings show.

Interventions on latent thoughts. Chang et al. (2026) modify cached latent thoughts during inference, without any training, pushing them toward directions associated with correct reasoning and improving accuracy across scales. Their causal analysis finds that early thought vectors carry the largest causal influence on the answer. While they steer entire thoughts to improve accuracy, we look inside the thought to identify what it carries. Li et al. (2026) apply causal interventions to COCONUT and CODI one step at a time, discovering that different latent steps have different roles and that an intervention’s effects cannot be observed until much later steps. They do not look inside a single step or study the structure/identity split, but their findings are consistent with our transplant results. To our knowledge, this paper is the first causal test to determine the internal composition of continuous thoughts.

Readable is not causal. A known blind spot of probing is that readable information is not necessarily used in computation (Belinkov, 2022; Hewitt and Liang, 2019). Techniques such as iterative null-space projection (INLP) (Ravfogel et al., 2020) and amnesic probing (Elazar et al., 2021) aim to address this causal gap. We apply these methods, including the control that removes an equal number of random directions, to latent CoT, and the result turns out to be one of the most striking instances of the read/use asymmetry: we find near-perfect decodability, but no behavioral effect. Othello-GPT (Li et al., 2023a; Nanda et al., 2023) is a contrast to our paper, as it has a similar read/rewrite arc, but with the opposite second part: their linearly decoded state was also linearly editable. This comparison shows that readable states cannot be trusted as steering mechanisms, and our substitution experiments are continuations that their results never made necessary.

Causal-intervention methodology. The methods we applied included activation patching (Zhang and Nanda, 2024; Meng et al., 2022; Wang et al., 2023), as our transplants are interchange interventions whose donor construction, controls, and non-candidate probabilities are adapted to latent CoT. Our matched donor design is based on activation steering (Turner et al., 2023; Li et al., 2023b; Rimsky et al., 2024), with both techniques built on difference vectors. The causal Jacobian basis was adapted from Gurnee et al. (2026) to align with latent reasoning (§5.1, Appendix E). To our knowl-

edge, SAE feature intervention is applied only for discrete reasoning (Chen et al., 2025), and dictionaries for latent thought have not been explored yet, supporting the choice of probes in our methodology.

Limitations

The scale of the model. We restrict our analysis to a model of limited size because claims about the search state require frontiers, paths, and reachability to be identifiable. This experimental setup is one in which these can be extracted. For natural tasks, the operations described in Sections 4–6 cannot even be defined. We test the hypothesis in its original environment, with the same methodology, by adopting its architecture, task, and scale. Additionally, Rizvi-Martel et al. (2026) observed that representations in superposition are distinguishable only in small models trained from scratch, because fine-tuned larger models find more efficient shortcuts and do not usually make use of their latents. Scale up is therefore undefined rather than deferred, especially considering there is no known larger setting for which the hypothesis is live, as a larger model would have bypassed the latent thought’s components instead of using them.

Limits of the linear representation. After editing the nodes under the pre-registered bases, an exploratory causal-Jacobian basis, and full linear-subspace removal (INLP), we find that per-branch linear edits have no causal effect and reject the sum-of-branches reading. We do not, however, rule out nonlinear encodings, as the construction of Zhu et al. (2025b) is only an existence proof, and trained models may implement the frontier in a way that no linear dictionary can express. The causal Jacobian basis aligns weakly with node directions mid-reasoning and strongly at the readout (0.5 versus 0.93, Appendix E), consistent with a mid-reasoning thought that is causal but not node-identity dependent. Therefore, nonlinear readings are a natural next step.

Training reproducibility. Accuracy on held-out problems is influenced by the training seed, falling within 90.2–95.7. Based on the pre-specified criteria, two out of four seeds fail to reach the pass threshold. However, the random seed affects only the model’s ability to convert the search into the correct answer, not the hidden search state map, which is constructed correctly in each seed. To

keep the set bias-free, we did not select any new seeds after observing the outcomes.

Generality across depth. All of our empirical findings are based on a single model with the depth locked in at two layers, across four seeds. According to Rizvi-Martel et al. (2026), four layers is the maximum depth where the pattern persists, so replication at greater depth is a natural next step, with the pipeline unchanged.

What carriage does and does not mean. Single-step transplants are mostly recovered from the prompt (−2.9 versus sustained −91.7), so carrier refers to the sustained replacement’s dominant channel, not the only place the search is held. In a small number of graphs, identity-targeted label swaps leave behind minor traces of identity within the structural map (minimum cosine similarity 0.76, 2% of transplants flip, and interior perturbations to 0.77 move behavior by only −0.08). While the structure-identity split is overwhelmingly strong, it is not absolute.

Future work. Non-linear readings of the search states have not been explored in this study, and which dictionary, if any, expresses the carried structure remains an open question. Because the pipeline does not depend on the basis, cross-family transplants are possible (for example, CODI or recurrent-depth models). Whether the structure/identity split holds in the same manner is an explicit test of generality. Finally, because token identity is bound only at readout, future monitoring and steering interventions should specifically target these binding zones instead of the intermediate components (§5.4).

9 Conclusion

What is actually inside a continuous thought? The search frontier is almost perfectly readable, but its content has no causal effect, as per-branch edits in any pre-specified basis, even under full linear-subspace removal, leave the answer unchanged. However, the thought is not just a passive byproduct: substituting it entirely redirects the answer to the donor’s when the donor differs in structure, and does nothing when it differs only in which candidate is correct. Structure is carried, while identity is bound late.

Reading out latent CoT reveals a precise but non-causal channel. Readouts remain useful as observations, but our findings show they cannot be

trusted as steering mechanisms. If identity is written only at readout, then steering belongs at the final, binding step. The final-step swap, which flips the answer completely, is proof that such zones can be located and used. Because an observational pattern can still fail as a mechanism, theoretical models of superposition now have a causal threshold to pass.

In this experimental setup, the thought ended up being neither an editable sum of branches nor a neutral byproduct, but a third thing: a carrier of search structure that computes its true identities late. Finding the dictionary that can express it is now a clear frontier.

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A Pre-registration documents and decision criteria

We present two time-stamped analysis plans that were frozen before corresponding data existed (METIS-1 2026-07-07, METIS-2 2026-07-08). A frozen document strictly defines its confirmatory

analyses, and each later change is documented and reported as an amendment (App B). These documents use the project’s internal jargon ("gate", "bar", "KILL"), but the mapping to this paper’s terms is Table 15 (App F).

A.1 METIS-1: thought-arithmetic rejection test (frozen 2026-07-07)

The following frozen document (commit hash [7a8b9c0]) pre-registers the decision criteria for the branch arithmetic tests.

2. Setup (FROZEN)

- **S1 (verdict substrate):** 2-layer GPT-2-style decoder (d_model=768, 8 heads) trained from scratch per arXiv:2505.12514 (AdamW $\beta = (0.9, 0.95)$, weight decay $1e-2$, constant LR $1e-4$; COCONUT multi-stage curriculum with previous-stage mixing $p=0.1$; starting from the authors’ released code, github.com/Ber666/reasoning-by-superposition). Training deviation from the paper’s fixed 25-epochs/stage schedule: per-stage early stopping on validation-accuracy plateau (patience 3, max 25 epochs/stage), cap ~ 36 h wall-clock on M1/MPS — acceptable because the substrate precondition below gates acceptance on outcomes, not schedule. A transfer arm to a larger model was registered as non-gating and later retired (App B).
- **Task (verified from the paper, not assumed):** ProsQA-style binary forced choice. Graph edges in the prompt; two candidates c_1, c_2 ; exactly one reachable from root r (the decoy is globally unreachable). Every node is a dedicated vocabulary token.
- **Substrate precondition — no gate data may be collected until both hold on the primary seed:** (a) held-out test accuracy $\geq 93\%$ (paper reports “near-perfect”, Fig. 4 ≈ 0.97 – 1.0 ; no exact number printed); (b) the BFS-wave inner-product readout replicates (Fig. 6 / Table 5 pattern: Frontier/Optimal $\gg$ Reachable $\gg$ Not-Reachable ≈ 0 , sweeping outward per step). Precondition failure means fix training, never gates.
- **Eval graphs:** generated from the training-distribution generator with constraints: at intervention step t the reachable frontier

contains exactly 3 nodes (branching 3) with disjoint downstream subtrees (unique endpoint $\rightarrow$ branch attribution); the answer candidate c_{i^*} lies in exactly one branch’s subtree; the decoy is unreachable as in training. $N = 500$ held-out graphs per condition, fixed seed list.

- **Measurement:** $T = \text{softmax probability of the correct candidate } c_{i^*} \text{ at the answer position, renormalized over } \{c_1, c_2\}$. Chance floor = 50%. Escape mass $1 - (p(c_1) + p(c_2))$ under the full softmax is reported per condition (a large increase indicates degeneration, not clean editing).
- **Basis:** node input-embeddings u_v (S1’s natural basis; one token per node). Pre-declared fallback ladder if the basis fails sanity: unembedding rows, then probe directions — each swap is a numbered DECISIONS entry.
- **Sanity (frozen definition):** on 100 training-distribution graphs, a basis passes iff answer-branch SUBTRACT yields median $\Delta T \leq -25$ pts AND its matched random-direction control yields median $|\Delta T| \leq 10$ pts. Gates are evaluated on the FIRST basis that passes sanity, chosen before any gate data is seen. If no basis passes sanity, that is Gate A failure.
- **Interventions** — applied once to the recycled continuous-thought vector at step $t =$ the ground-truth BFS depth at which the target branch’s root enters the frontier (known exactly from the synthetic graph; inner-product-readout-based step selection is reported as a robustness check only). Subsequent steps run free. Thoughts are unit-norm under the paper’s $\text{LayerNorm}(x) = x/\|x\|_2$, so all edited vectors are renormalized to unit norm:
 - SUBTRACT: $t' = \text{normalize}(t - \langle t, \hat{u}_v \rangle \hat{u}_v)$.
 - COLLAPSE: $t' = \text{sign}(\langle t, \hat{u}_v \rangle) \cdot \hat{u}_v$ — the pure branch direction; equivalent to a size-1 frontier state, hence near-in-distribution.
 - INJECT: $t' = \text{normalize}(t + \gamma \cdot \hat{u}_x)$, $\gamma =$ per-graph median $\langle t, \hat{u}_f \rangle$ over true frontier nodes f . Exploratory only.
 - **Controls, matched per intervention:** (gating) the same operation with a ran-

dom unit direction, matched magnitude; (reported, non-gating) the same operation with a uniformly sampled non-frontier node’s embedding, matched magnitude.

- **Seeds:** gates are evaluated on the primary training seed. Two additional seeds are trained and reported as robustness (non-gating), only after the primary-seed verdict is locked and only if budget allows.

3. Metrics and gates (FROZEN)

Definitions: per eval graph, T as in §2. “Answer branch” = the frontier branch at step t whose subtree contains c_i^* . “Non-answer branch” = one of the other two frontier branches, uniformly sampled; its subtree contains neither candidate.

- **Gate A¹ (subtraction is causal & selective) — all three or fail:**

- (i) SUBTRACT answer branch: median $T_{\text{post}} \leq 60\%$ AND median $\Delta T \leq -30$ pts.
- (ii) SUBTRACT non-answer branch: median $|\Delta T| \leq 10$ pts (a surgical edit to an irrelevant branch must not disturb the answer).
- (iii) Matched random-direction control of (i): median $|\Delta T| \leq 10$ pts.
- If (iii) fails, the reported conclusion is “arithmetic indistinguishable from generic fragility,” not “subtraction has no effect.”

- **Gate B (collapse commits) — all three or fail; strengthens the claim, cannot flip the verdict in either direction:**

- (i) COLLAPSE onto answer branch: median $T_{\text{post}} \geq 80\%$ (the branch component alone must carry the answer through re-expansion).
- (ii) COLLAPSE onto non-answer branch: median $T_{\text{post}} \leq 60\%$.
- (iii) Random-direction collapse control: median $T_{\text{post}} \leq 60\%$, AND (i)’s median T_{post} exceeds this control’s median T_{post} by ≥ 20 pts.

- The subsequent-step inner-product readout (does the collapsed-onto branch’s subtree, and only it, expand at steps $> t$?) is reported, non-gating.

- **Verdict:** Gate A pass $\rightarrow$ the paper’s core claim stands regardless of B (B strengthens it). Gate A fail on S1 after basis fallbacks $\rightarrow$ **KILL:** report “continuous thoughts are not causally-linearly editable” as the finding. No rescue, no post-hoc bars.

- **INJECT and the capacity curve (branching 2..8): EXPLORATORY.** Reported always; cannot flip a verdict. Sharp exploratory prediction from the paper’s answer-head construction: INJECT of $\gamma \cdot u_{\text{decoy}}$ at step t should raise the decoy’s share. Branching > 4 pushes graph size beyond the training distribution (~ 23 nodes) — off-distribution confound, reported alongside the curve.

- Sample-size rule: any gate quantity within 5 points of its threshold $\rightarrow$ double N once to 1000, decide on pooled. (Explicitly scoped to THIS section’s gates only — lesson of mycelia DECISIONS #007.)

A.2 METIS-2: the carrier code — minimal-pair transplant test (frozen 2026-07-08)

The following frozen document (commit hash [d9e2f3a]) pre-registers the decision criteria testing the candidate-resolution of the carried state.

2. Setup (FROZEN)

- **Substrate:** primary model² (2-layer, $d=768$, 8 heads; substrate precondition PASSED 2026-07-08: test 94.51% $\geq$ 93%, BFS-wave readout at all steps — results/s1_seed0_win/acceptance.json). Checkpoint ckpts/s1_seed0_win/best.pt, device cuda, seed 0. **Seeds 1 and 2:** retrained with the identical recipe (fixed 25-epoch stages 0–3, full-task patience 15); each must pass the same precondition to be reported; robustness only, non-gating, run after the primary-seed verdict is locked.
- **Recipients:** evaluation set ($n = 100$)³ (the arms-1–3 slice; escalation set train[:300]).

²Repository identifier: s1_seed0_win

³Repository identifier: train[:100]. The escalation set is train[:300].

¹Gate = confirmatory analysis; see Table 15.

- **Minimal-pair construction (label surgery; donors always constructible):**

- *Candidate-swap donor (Gate A)*: recipient graph G with tokens T (target) and D (decoy) transposed in the edge list and reachability matrix⁴; root unchanged; the donor sample’s (target, neg_target) FIELDS are set to the recipient’s so the question and root lines serialize identically under the same pinned seed. Net textual difference between donor and recipient prompts: only the swapped token occurrences inside edge tuples. Semantics: D reachable exactly where T was; answer flips by construction; on-distribution (ProsQA graphs are label-symmetric).
 - *Placebo donor (Gate A control)*: the two lowest-id nodes that are unreachable from root and $\notin \{T, D, \text{root}\}$, transposed the same way. Candidate reachability and the answer are unchanged; textual edit is of the same kind.
 - *Path-interior donor (reported, non-gating)*: lowest-id node in $\text{neighbor_k}[1..K-1]$ transposed with the lowest-id unreachable non-candidate node — same search shape, different interior identity; probes whether interior node identity is part of the carried state.
 - Guards: skip and count recipients where a required pair does not exist.
- **Transplant operation (as in #010)**: capture the donor run’s recycled thoughts under the donor’s pinned serialization; inject AS-IS (natural norm ~ 27.4) at all intermediate passes $0..K-2$; the final thought is computed naturally downstream.
 - **Mixing operation (Gate B)**: at each intermediate pass, $t_{\text{mix}} = (1-\alpha) \cdot t_{\text{own}} + \alpha \cdot t_{\text{donor}}$, rescaled to $\|t_{\text{own}}\|$ (on-manifold per #005); $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$; donor = the candidate-swap donor.
 - **Measurement**: T , escape, medians, answer position — identical to PREREGISTRATION.md §2. Serialization pinning per DECISIONS #008 (per-graph seed $(\text{gi} \ll 16)^{\text{seed}}$, edges restored before every forward) is part

of the frozen protocol. One run per condition; no iteration on any downstream number. **$n = 100$ is a frozen floor for every gated statistic** (per-graph effects are bimodal — smoke discovery 2026-07-08 — small- n medians whipsaw); smoke runs are plumbing checks BY DESIGN and carry no evidential weight on effect sizes.

- **Placebo scope (stated, per external review R3)**: the placebo transposes two UNREACHABLE non-candidates, whose mid-search readout weight is ≈ 0 (acceptance.json) — so Gate A’s placebo controls for edit type and serialization, NOT for state-difference magnitude. The path-interior swap is the informative middle condition (same shape, different interior identity) and is deliberately non-gating: a nonzero interior effect is a finding about WHAT is carried, not a failed control.
- **Controls run alongside (all gating unless marked)**: baseline; reserialized baseline (noise floor); self-transplant (must be exactly 0 — plumbing); per-step zero (health check; expect ≈ -43); placebo transplant; path-interior transplant (reported).

3. Metrics and gates (FROZEN)

- **Gate A (the carried state is candidate-resolved) — all five or fail:**
 - (i) candidate-swap transplant: median $T_{\text{post}} \leq 60$ AND median $\Delta T \leq -30$ AND median escape ≤ 0.05 (external review R1: the escape ceiling separates REDIRECTION from representation DESTRUCTION — the confound #010’s random-donor control exposed at $-30.4 \Delta T / 0.98$ escape; clean carrier effects measured two orders below the ceiling);
 - (ii) fraction of recipients with $\Delta T \leq -50$ is ≥ 0.5 , and the full per-graph ΔT histogram is reported (external review R2: per-graph effects are bimodal, so a median alone can mask a large untouched minority);
 - (iii) placebo transplant: median $|\Delta T| \leq 10$;
 - (iv) self-transplant: median $|\Delta T| \leq 0.01$ pp (fp-realistic form of “exactly zero”;

⁴Repository identifier: neighbor_k

failure voids the run — fix plumbing, never numbers);

- (v) per-step zero: median $\Delta T \leq -30$ (instrument sensitivity).
- **Gate B (the carrier is continuous/graded)**
— all three or fail; cannot flip Gate A:
 - (i) medians of ΔT over the α grid are nonincreasing within a +3-pt tolerance per step AND span a total range ≥ 30 pts;
 - (ii) the $\alpha = 1$ endpoint reproduces Gate A(i)’s median ΔT within 10 pts;
 - (iii) median escape at $\alpha = 1 \leq 0.05$. Per- α escape medians are REPORTED at every grid point; PRE-COMMITTED interpretation: elevated escape at intermediate α (median > 0.2) means own/donor mixtures leave the thought manifold — a finding about the code’s geometry, not a gate failure.
 - **Curve-shape taxonomy (pre-committed, external review R6; reported, non-gating):** per graph whose own $|\Delta T|$ range ≥ 30 , the curve is “snap” if $\geq 80\%$ of that graph’s range is traversed in a single α increment, else “graded”; graphs with range < 30 are “unresponsive”. The three proportions are reported; all outcomes are findings.
- **Verdict semantics:** Gate A pass $\rightarrow$ claim “S1’s mid-search thought state causally carries candidate-resolved search information, transplantable across single-label counterfactuals” stands; Gate B refines it toward a continuous state variable. Gate A fail $\rightarrow$ METIS-2’s core claim is dead: #010’s carriage was coarse (whole-graph state), not candidate-resolved — reported as the finding, no rescue, no post-hoc bars.
- **Sample-size rule:** any gated pp-scale quantity within 5 pts of its threshold, or the Gate A(ii) fraction within 0.05 of 0.5 $\rightarrow$ escalate once to train[:300], decide on the pooled set. Escape ceilings are excluded from the escalation rule (different scale; the ceiling is generous by two orders of magnitude). Scoped to this section only.

4. Registered pre-freeze extensions (non-gating)

- **Test-split replication (R4):** Gate A conditions rerun on the first 100 test-split graphs (never seen in training; serialization seed namespace $((gi + 1000000) \ll 16) \wedge$ seed to avoid collision with training-graph seeds). Removes the memorized-graphs objection; sub-hour.
- **INLP-style subspace removal (amnesic-probing objection):** the kill-test’s probe rung removed ONE direction per node; the amnesic-probing literature shows single-direction removal can under-delete a concept. A small iterative nullspace-projection arm per node (fit on the #006 recipe’s split) either pre-empts the strongest reviewer objection to the KILL or bounds it; the METIS-1 verdict is scoped to its frozen ladder and is not at stake.

A.3 Evaluation-set constraints

- **Eval graphs:** generated from the training-distribution generator with constraints: at intervention step t the reachable frontier contains exactly 3 nodes (branching 3) with disjoint downstream subtrees (unique endpoint $\rightarrow$ branch attribution); the answer candidate c_{i^*} lies in exactly one branch’s subtree; the decoy is unreachable as in training.
- **Measurement:** $T = \text{softmax probability of the correct candidate } c_{i^*} \text{ at the answer position, renormalized over } \{c_1, c_2\}$. Chance floor = 50%. Escape mass $1 - (p(c_1) + p(c_2))$ under the full softmax is reported per condition (a large increase indicates degeneration, not clean editing). **$n = 100$ is a frozen floor for every gated statistic** (per-graph effects are bimodal — smoke discovery 2026-07-08 — small- n medians whipsaw); smoke runs are plumbing checks BY DESIGN and carry no evidential weight on effect sizes.
- **Sample-size rule:** any gate quantity within 5 points of its threshold (or the Gate A(ii) fraction within 0.05 of 0.5) $\rightarrow$ double N once to 1000 (or escalate once to train[:300]), decide on pooled. Escape ceilings are excluded from the escalation rule (different scale; the ceiling is generous by two orders of magnitude). (Explicitly scoped to THIS section’s gates only).

ID	Date	Status	What changed	Criteria touched?
#001–#002	2026-07-06/07	Pre-freeze	Draft-stage history, metric formulation, and spec fixes.	(Pre-freeze)
#003	2026-07-07	Post-freeze	Early-stopping relaxation, full-task phase (training op).	No
#004	2026-07-07	Post-freeze	Precondition-miss retrain recipe (withdrawn run kept, labeled).	No
#005	2026-07-08	Post-freeze	Renormalization target correction (unit norm to pre-edit norm).	No
#006	2026-07-08	Post-freeze	Basis-fallback ladder rung-1 swap (unit-norm variant retired).	No
#008	2026-07-08	Post-verdict	Serialization noise found → pinning (App C).	No
#011	2026-07-08	Freeze	METIS-2 freeze (user bar review).	No
#012	2026-07-08	Post-freeze	S2 / COCONUT-GPT-2 transfer arm RETIRED.	No
#013	2026-07-08/09	Post-freeze	seed-2/seed-3 precondition misses → registered fallback; no-further-seeds guard.	No
#014	2026-07-09	Post-freeze	INLP recipe pinned; $k \leq 8$ under-null; full-config note.	No

Table 9: Protocol amendments transcribed from the project’s internal log. A “No” in the final column indicates that the pre-registered thresholds and evaluation rules governing the confirmatory analyses remained strictly untouched.

B Protocol amendments

Table 9 reports the complete amendment log. No entry amended pre-registered criteria.

C Serialization determinism and the reference-pipeline fix

C.1 An unseeded shuffle in the reference pipeline

During the escalated deletion stage, we noticed that the zeroing sensitivity control, a value that had replicated across earlier tests, failed to reproduce between identical runs. This discrepancy was caused by the data pipeline, not by the implementation.

The reference pipeline (expand_data/get_prefix in dataset.py of the public code) shuffles each problem’s edge list *in place* using Python’s random global, unseeded module, so the candidate order is randomly decided on every call. Therefore, every forward pass sees a new serialization of the same graph, so no two runs are guaranteed to have the same prompt token order for the same problem. Because this function shuffles the edge list in place, seeding the generator would not restore determinism, as the lists were already perturbed. The original order must be restored before each forward pass.

Our fix is adopted as an amendment (App. B) and it is based on executing three operations before every condition of a problem: restoring the original edge order, seeding the generator with a value assigned to each problem ($(g_i \ll 16) \oplus s$ for problem index g_i and run seed s), and guaranteeing that all conditions of a problem share one serialization and

Condition (input-embedding basis, $n = 100$)	unpinned	pinned
per-step zeroing (sensitivity control)	−51.5	−42.9
delete answer-carrying path, incl. final step	−33.2	−65.1
delete answer-carrying path, excl. final step	−0.01	−0.01
one-shot / per-step SUBTRACT, random controls	−0.0	−0.0

Table 10: The same analysis under the reference pipeline’s unseeded serialization versus pinned serialization (median ΔT). The null conditions are not affected, but the large-effect conditions get substantially modified under the unseeded shuffle. The pinned values reproduce exactly on re-runs, and the general conclusion is unchanged in both columns.

that runs are dependent only on the process.

A re-serialization condition (baseline for a new serialization seed) was added to every subsequent test as the minimum acceptable value for measured noise. Its value is -0.0 median ΔT . The already-registered analysis runner was left the same on purpose, as its recorded values provide serialization variance as a documented measurement caveat, and the resulting files from the affected exploratory test were re-run after pinning. The original runs were documented and released under an explicit unpinnedserial label instead of being overwritten.

Every null condition is a null in both columns, and an effect affecting the qualification state would have been detected through it. The effect of the path deletion was first observed through this noise. For the fix, the matched-donor and label-swap donor constructions of Sec. 6 require the prompts

to differ only in the targeted content, not by random serialization, which the bug initially caused, corrupting every pair.

C.2 Bit-exact reimplementaion of the reference code

The code for reference training is hard-wired to two-GPU distributed execution. The wall-clock cost is dominated by tens of thousands of per-element device synchronizations for each training step, not necessarily by model size. Through reimplementaion, we introduce a vectorized, single-process pipeline, bringing training to consumer hardware (hours on a single laptop GPU).

The equivalence is established by a released test run (`tests/test_fast_equivalence.py`) on real training data at the deepest stage, on CPU in evaluation mode. The test requires identical tokenization on 200 real problems, an identical forward loss with a maximum logit difference of exactly 0.0, and identical gradients for every parameter.

Passing this test was required, and it was re-run on each machine before any training on that machine was used. All experiments in this paper run on the re-implementaion, and the reference only acts as a baseline to verify computational equivalence.

D Full results: distributions, seeds, robustness

D.1 Per-graph distributions

The intervention effects on graphs are not normally distributed, so the pre-registered statistics are a median plus the fraction of graphs with $\Delta T \leq -50$ instead of a mean. For the label-swap transplant, the distribution is mostly concentrated around zero, as on the primary model, 78 of 100 graphs land in $[-10, 0)$, 18 at ≈ 0 , and only 2 flip fully.

For the zeroing, the distribution is much wider and concentrated at the two extremes: 40 graphs move by less than 10 points, 27 collapse by 90 or more, and the remaining 33 spread between.

Neither distribution would be accurately represented by the mean, and the median alone would hide the flipped minority, so we choose to employ the paired statistics.

D.2 The four training runs

Table 11 reports all four training runs with the zeroing baselines and extends Table 2.

D.3 Replications of the label-swap test, in full

Table 12 reports the complete sensitivity control tests for both replications of the label-swap transplant test, reflecting the primary run’s set of tests in the main text (Table 6).

Held-out histogram detail: 55 of 100 graphs in $[-10, 0)$, 27 at ≈ 0 , and 8 graphs at $\Delta T \leq -50$, with the flipped minority visible in Fig. 4’s third panel.

D.4 Dose-response in full

Table 13 shows the full α grid for the blending analysis of §6.2.

D.5 Accounting

For four of the 100 recipients in the transplant of entire thoughts analysis, a matched donor cannot be constructed, so they were skipped. This is why we only have $n=96$ rows in Table 5. In the label-swap test itself, there were no skips in any condition (candidate, placebo, and interior donors could be constructed for all 100 problems in all three runs).

E Basis and INLP details

E.1 The probe basis

We fit a logistic probe (Eq. 5) for each node v on step-1 thoughts. The label 1 is preserved only if v is a part of the depth-1 frontier of the problem’s graph. Probes are fitted on all 14,785 training problems, excluding the last 500 that are used only for evaluation. Across the 23 probed nodes, the median held-out AUC is 0.9979 (minimum 0.63). The unit-normalized probe weight vectors are the directions $\hat{\mathbf{u}}_v$ of the probe basis in Sec. 5, and this design was committed before the probes were fitted (App. B).

E.2 The causal-Jacobian basis

The exploratory basis of Sec. 5.1 is based on causal influence rather than geometry or readability. For each latent step, we compute the Jacobians of the candidate readout in relation to the thought vector on 2,000 training problems (problems 500–2500, separated from the evaluation set), and extract up to four main directions per step from the gradients each problem offers. Table 14 reports the architecture of these directions.

E.3 INLP configuration and trajectories

The under-deletion exploratory analysis (Sec. 5.3) has the exact probe design of §E.1 for each node:

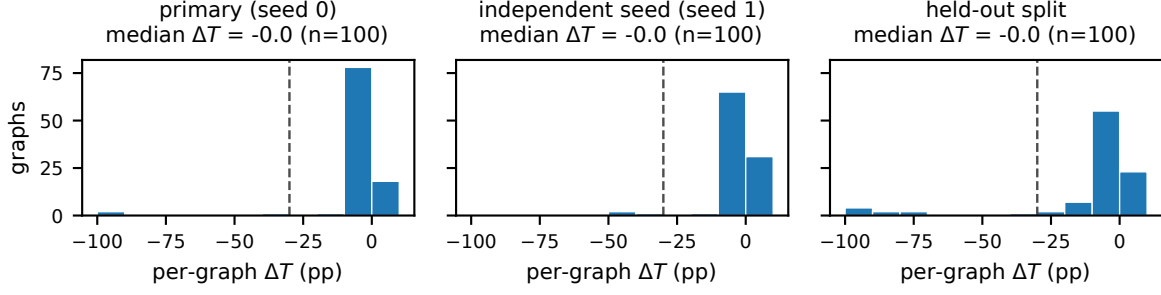


Figure 4: Per-graph effect of the label-swap transplant across all three runs: primary model, independent seed, and held-out split. The null replicates in each, and the held-out split shows the small flipped minority that its pre-specified sample-size flag refers to (Sec. D.3).

Model	Held-out acc.	Validity ($\geq 93\%$)	Ordering (Eq. 4)	Zeroing anchor
seed 0 (primary)	94.5	pass	all K steps	-42.9
seed 1	95.7	pass	all K steps	-36.8
seed 2	90.2	fail	all K steps	—
seed 3	92.8	fail	all K steps	—

Table 11: All four independently trained models. The structural readout ordering remains consistent at every latent step in every model, but the answer accuracy varies by seed. The two models that did not pass were excluded before any intervention was run on them, so we do not have zeroing baselines for them, as the exclusion happened because they were not valid, not after observing the intervention outcomes.

	primary	ind. seed	held-out split
median ΔT (candidate swap)	-0.0	-0.0	-0.01
median T_{post}	99.98	99.98	99.91
median non-candidate probability e	0.0001	0.0001	0.0002
fraction $\Delta T \leq -50$	0.02	0.0	0.08
path-interior swap (reported)	-0.08	-0.31	-0.12
placebo swap, median $ \Delta T $	0.0	0.0	0.0
self-transplant	0.0	-0.0	-0.0
re-serialization noise floor	0.0	0.0	0.0
per-step zeroing (sensitivity)	-42.9	-36.8	-25.4

Table 12: The full battery for both replications. The null and every control replicate. The held-out split’s sensitivity baseline is not as strong (-25.4), and its closeness to the -30 reference value triggered the pre-specified sample-size flag on the instrument-sensitivity margin, not on the candidate result, which greatly differs from every threshold in all three runs. The held-out replication was registered as non-binding.

α	0	0.25	0.5	0.75	1
median ΔT	0.0	-0.0	-0.0	-0.0	-0.0
median e	0.0001	0.0001	0.0001	0.0001	0.0001

Table 13: Blending recipient and donor thoughts at mixing weight α (norm-preserving). The curve is flat, as entailed by the behaviorally identical extremes. Blends stay on-distribution at every α , so there is no elevated non-candidate probability. The $\alpha=1$ endpoint reproduces the transplant result exactly (difference 0.0), and it constitutes a cross-consistency check for the two runners.

we fit a new probe on the projected thoughts, re-

move its direction, and repeat until the held-out AUC drops to 0.55 or an iteration limit of 40 is reached. Fifteen of the 23 nodes hit the iteration limit, while the others stop between $k = 2$ and $k = 34$, and the median is $k = 40$. Figure 5 shows the per-node trajectories.

The deletion’s results coincide with the ones from the qualification run: on the same evaluation problems, the answer node’s directions are projected out of the step-1 thought, and the edited vector is rescaled to its pre-edit norm. We compare the following: are removing the single first probe direction (which reproduces the qualification run’s value), fully removing the subspace

latent step	1	2	3	4
cross-graph concentration	0.50	0.49	0.71	0.93
median $ \cos $ to token embeddings	0.41	0.55	0.77	0.78
median $ \cos $ to probe directions	0.03	0.05	0.13	0.13

Table 14: Geometry of the causal-Jacobian directions by latent step. Cross-graph concentration measures how consistently the per-problem causal directions align across problems, and mid-reasoning they are much more dependent on the problem they are operating on (≈ 0.5), and they tend to concentrate in a globally consistent direction (0.93) only at the final step. The directions are genuinely new, nearly orthogonal to the probe directions everywhere, and rotating toward the token-embedding geometry only at the end.

($\tilde{\mathbf{z}} = \|\mathbf{z}\| \cdot \text{normalize}((\mathbf{I} - \mathbf{B}\mathbf{B}^\top)\mathbf{z})$ for the node’s orthonormal direction matrix $\mathbf{B}$), and removing an equal number of random orthonormal directions (the rank-matched control). Results are reported in Sec. 5.3: full-subspace removal -0.0 (with $T_{\text{post}} = 99.99$), rank-matched random removal 0.0 .

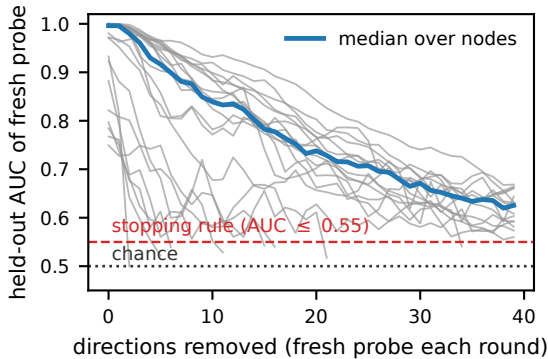


Figure 5: INLP removal trajectories, one per node: held-out AUC of a newly fitted probe after each projection. Most nodes hit the iteration limit with the remaining signal near but above chance (final AUC 0.5–0.67), and this redundancy is what motivates full-subspace removal (Sec. 5.3).

F Repository mapping and claims-to-evidence

The released repository acts as the research project’s laboratory record, so the project’s internal working vocabulary is used. Table 15 maps it to the paper’s terms so that the registered documents reproduced in Appendix A can be read against the main text.

Table 17 is an extension of the claims-to-evidence table of the main text (Table 16).

Repository term	This paper
METIS-1, METIS-2	confirmatory tests 1 (Sec. 5) and 2 (Sec. 6.2)
frozen	registered; fixed by a named commit before data
bar	pre-specified threshold
gate; Gate A/B	confirmatory analysis and its criteria set
KILL	pre-specified rejection
(basis) sanity	qualification threshold (Table 3)
substrate precondition	validity criterion (Table 2)
escape (mass)	non-candidate answer probability e (§3.1)
health check (per-step zero)	positive sensitivity control
smoke run	pilot plumbing check; carries no evidential weight
DECISIONS.md	amendment log entries (App. B)
#nnn	
LOG.md	append-only laboratory notebook
S1; s1_seed0_win	the substrate; the primary model
S2	a registered transfer arm to COCONUT-GPT-2, retired (App. B)
train[:100]	/ evaluation set / escalation set
train[:300]	
neighbor_k	depth- k valid-path node sets (the “optimal” category)
arm #010	the whole-thought transplant exploration (Sec. 6.1)

Table 15: Mapping from the repository’s internal vocabulary to this paper’s terminology.

All numeric values in the paper are macros generated by scripts/make_numbers.py directly from the project’s files. There is one exception to file-level provenance: the donor self-consistency figures that appear in Sec. 6.2 (median $T(\text{answer}=D) = 99.8$, 79% correct) are sourced from the repository’s record of the validation diagnostic instead of a released result file.

Claim	Analysis	Headline number	Source files
frontier readable	probes (4.2)	0.9979	probe_basis_report
thought necessary	ablation (4.3)	−42.9	explore_perstep_*
per-branch edits inert	basis seq. (5.2)	~0 all cells	basis_sanity*
not under-deletion	INLP (5.3)	~0 at full subspace	inlp_arm_*
identity editable at readout	swap (5.4)	−99.97	explore_jlens
thought carries state	transplant (6.1)	−91.7	explore_patch
carriage is structural	label-swap (6.2)	−0.0	m2_gateA_*
identity rides inert	cosines (6.3)	0.9975 vs 0.34	diag_m2_*

Table 16: Headline claims, the analyses that test them, and the files in our released repository containing the underlying results. Repository naming is mapped to the paper’s terminology in Appendix F.

Paper object	Repository source
Fig. 1 (three verbs)	scripts/fig_three_verbs.py (values read from source files)
Fig. 2, Table 2	acceptance.json (all four models); scripts/fig_wave.py
probe decodability (§4.2)	probe_basis_report.json
Table 3 (qualification)	basis_sanity_{unitnorm_v1, inputemb_normpreserving, probes}.json
Table 4 (escalated deletion)	explore_perstep_{inputemb, probes}.json;
	explore_jlens.json
INLP results (§5.3)	inlp_arm_summary_full.json; inlp_basis_report_full.json
final-step swap (§5.4)	explore_jlens.json
Table 5 (transplants)	explore_patch.json
Table 6, Fig. 3	m2_gateA_summary.json + _rows.jsonl (primary, seed-1, held-out)
dose-response (§6.2, App. D)	m2_gateB_summary.json
Table 7, residual analysis (§6.3)	diag_m2_structure.json; diag_m2_residual.json
serialization exhibit (App. C)	explore_perstep_inputemb_unpinnedserial.json
every numeric macro in this paper	scripts/make_numbers.py → numbers.tex
registered documents; amendments	PREREGISTRATION.md; PREREGISTRATION_M2.md; DECISIONS.md

Table 17: Where each reported result comes from. Repository: <https://github.com/star2vec/RRR>, release tag v1.0-arxiv.